

Linear Algebra Notes

FRORDO

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1. What is Linear Algebra?

We will not answer this question immediately, except to say that linear algebra is the study of *linear transformations*. Here are some definitions and axioms to remember:

Definitions

1. A vector is an element of a vector space
2. Vectors are denoted by small bold letters, say $\mathbf{v}$ and matrices by capitals, A .
3. An $m \times n$ matrix means m rows and n columns. We will deal with finite matrices.
4. Notation: $A = (a_{j,k})_{j=1,k=1}^{m,n}$. The entries of the matrix A will be denoted $[A]_{ij}$ or a_{ij} .

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

5. The transpose is defined by swapping rows and columns. $[A^T]_{ij} = [A]_{ij}$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Vector Spaces

Let $\mathbb{V}$ be a set. Let $\mathbf{v}, \mathbf{w}$ and $\mathbf{u}$ be arbitrary elements of $\mathbb{V}$, and a and b be scalars. $\mathbb{V}$ is called a *vector space* if it satisfies the following:

1. $+$ and $\times$ are functions from $\mathbb{V} \times \mathbb{V} \rightarrow \mathbb{V}$
2. **Commutative:** $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$.
3. **Associative:** $\mathbf{v} + (\mathbf{w} + \mathbf{u}) = (\mathbf{v} + \mathbf{w}) + \mathbf{u}$
 $(ab)\mathbf{v} = a(b\mathbf{v})$
4. **Additive identity:** There exists $\mathbf{0}$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$
5. **Additive Inverse:** There exists $\mathbf{w}$ such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$
6. **Multiplicative Identity:** $1\mathbf{v} = \mathbf{v}$
7. **Distributive:** $a(\mathbf{v} + \mathbf{w}) = a\mathbf{v} + a\mathbf{w}$
 $(a + b)(\mathbf{v}) = a\mathbf{v} + b\mathbf{v}$

Linear Transformation

Let $\mathbb{V}$ and $\mathbb{W}$ be vector spaces. A transformation $T : \mathbb{V} \rightarrow \mathbb{W}$ is called *linear* if:

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$
2. $T(a\mathbf{v}) = aT(\mathbf{v}) \forall \mathbf{v} \in \mathbb{V}$ and all scalars a .

$$\begin{bmatrix} 1 & 2 & \dots & 10 \\ 2 & 2 & \dots & 10 \\ \vdots & \vdots & \ddots & \vdots \\ 10 & 10 & \dots & 10 \end{bmatrix}$$

$$\left(\begin{array}{c|c|c|c} & & & \\ b_1 & b_2 & \cdots & b_n \\ & & & \end{array}\right)$$

A

A

$$\left(\begin{array}{c} \boxed{A} \end{array}\right)$$